



Lesson 1.9 — Piecewise Functions

Big idea: A piecewise function is one function with different rules on different pieces of its domain. Pick the right rule by checking which interval the input lives in.

Quick review

Read it like a menu: for each input, find the row whose condition is true. Use that row's formula.

At boundary points: use the rule whose condition includes that point (look for $\leq$ vs $<$).

Graphing: graph each piece on its own interval. Use closed dots for endpoints included, open dots for excluded.

Continuity check: if both pieces produce the same y-value at the boundary, the graph connects.

Worked example

Worked example. Let $f(x) = \{ 2x + 1, x < 3 ; x^2, x \geq 3 \}$.

$f(0)$ uses the top rule: $2(0) + 1 = 1$.

$f(3)$ uses the bottom rule (because of $\geq$): $3^2 = 9$.

At the boundary, the top piece approaches $2(3) + 1 = 7$, but the bottom piece gives 9. The graph jumps.

Warm-ups

1. Let $f(x) = \{ x + 2, x < 0 ; x^2, x \geq 0 \}$. Find $f(-3)$, $f(0)$, $f(2)$.
2. Let $g(x) = \{ 5, x \leq 1 ; 2x + 3, x > 1 \}$. Find $g(1)$, $g(4)$.
3. Sketch (in words) the piecewise $f(x) = \{ -x, x < 0 ; x, x \geq 0 \}$. What well-known function is this?

Practice

1. Let $f(x) = \{ 3x - 1, x < 2 ; -x + 7, x \geq 2 \}$. Is f continuous at $x = 2$? Why?
2. A parking lot charges \$4 for the first hour and \$2 for each additional hour, with a maximum of \$14/day. Write a piecewise function for the cost $C(t)$ where t is hours parked, $0 \leq t \leq 24$.
3. Graph (describe) $f(x) = \{ x + 1, x < -1 ; x^2, -1 \leq x \leq 1 ; -x + 2, x > 1 \}$. Find the domain and range.

Challenge

1. For what value of c is the function continuous at $x = 1$? $f(x) = \{ x^2 + c, x < 1 ; 4x - 5, x \geq 1 \}$.
2. A cell phone plan charges \$25/month for the first 2 GB of data, then \$10 per additional GB. Write the piecewise cost $C(g)$ for g GB and find $C(1.5)$, $C(2)$, and $C(5)$.



Answer key — Lesson 1.9

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $f(-3) = -1$, $f(0) = 0$, $f(2) = 4$.
2. $g(1) = 5$, $g(4) = 11$.
3. It's the absolute value function $f(x) = |x|$.

Practice

1. Top piece at $x=2$: 5. Bottom piece at $x=2$: 5. Same value $\rightarrow$ continuous.
2. $C(t) = \{ 4, 0 < t \leq 1 ; 4 + 2(t - 1), 1 < t \leq 6 ; 14, 6 < t \leq 24 \}$. (Cost caps when $4 + 2(t-1)$ reaches 14.)
3. Domain: all reals. Range: $[0, 1]$ from the middle piece, plus the linear pieces extend to $(-\infty, 0)$ on left and $(-\infty, 1)$ on right $\Rightarrow$ range $= (-\infty, 1]$.

Challenge

1. Match values at $x = 1$: $1 + c = -1 \Rightarrow c = -2$.
2. $C(g) = \{ 25, 0 \leq g \leq 2 ; 25 + 10(g - 2), g > 2 \}$. $C(1.5) = \$25$; $C(2) = \$25$; $C(5) = 25 + 10(3) = \$55$.